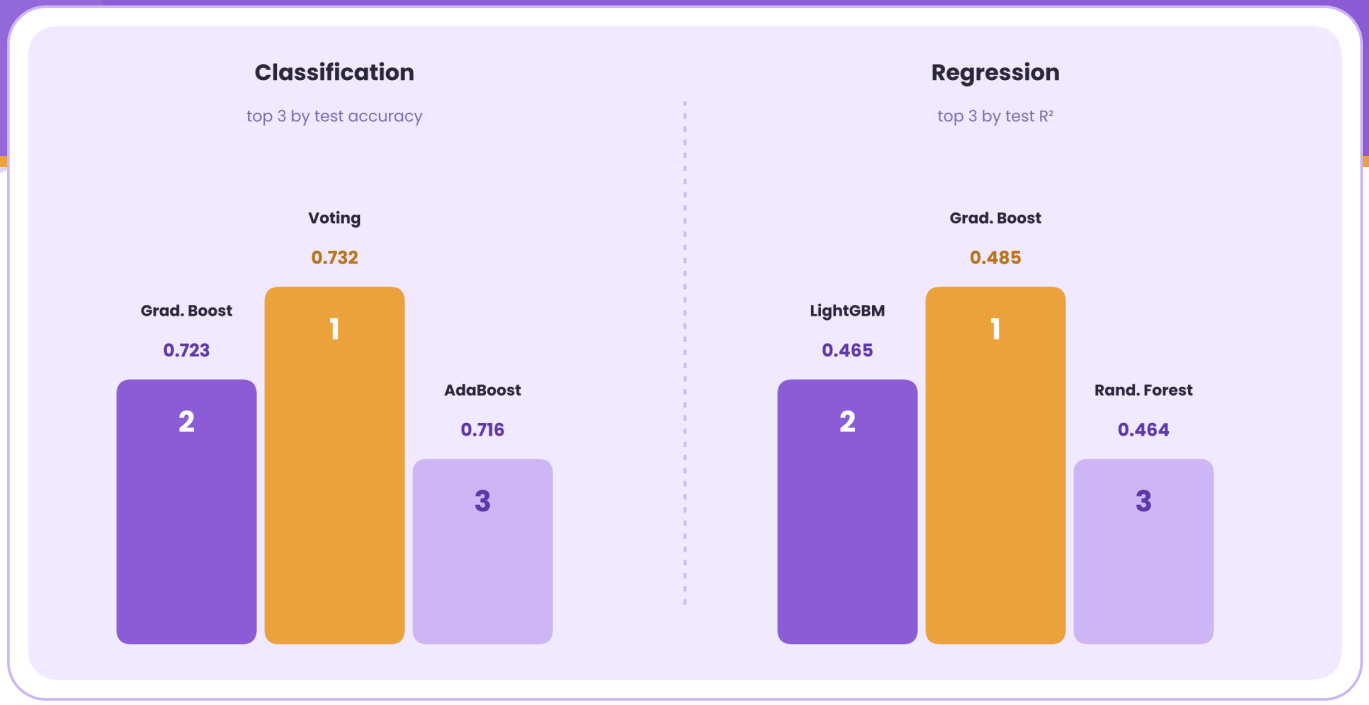
SUPERVISED MACHINE LEARNING

SMART OUTCOME PREDICTOR

Part G: Final Analysis & Reporting

Compare. Recommend. Deploy.

A clear read of what the notebook results show about bagging, boosting, voting and stacking, and which models to put into deployment.



5,200
students in the dataset

15
input features

80 / 20
train / test split (4,160 | 1,040)

01 Compare
Bagging vs boosting

02 Evaluate
Boosting algorithms

03 Deploy
Final recommendation

Q **Prepare a final report covering the points below.**

Every score here is a test-set result on 1,040 students. Classification is judged by accuracy and regression by R^2 score.

1 Impact of bagging vs boosting

Bagging fits many models in parallel on bootstrap samples and averages them, which chiefly cuts variance. Boosting fits models in sequence so each one repairs the last one's mistakes, which chiefly cuts bias.

Against a single model	One decision tree reached 0.645 accuracy, and bagging lifted that to 0.698, roughly 5 points higher.
Classification	AdaBoost (0.716), Gradient Boosting (0.723) and LightGBM (0.712) all finished ahead of bagging (0.698), while XGBoost (0.681) fell short.
Regression	Bagging (R^2 0.456) outscored AdaBoost (0.407) and XGBoost (0.390), but was itself beaten by Gradient Boosting (0.485) and LightGBM (0.465).
Overall	Gradient Boosting was the strongest bagging-or-boosting model on both tasks, yet boosting did not win automatically: some algorithms trailed bagging.

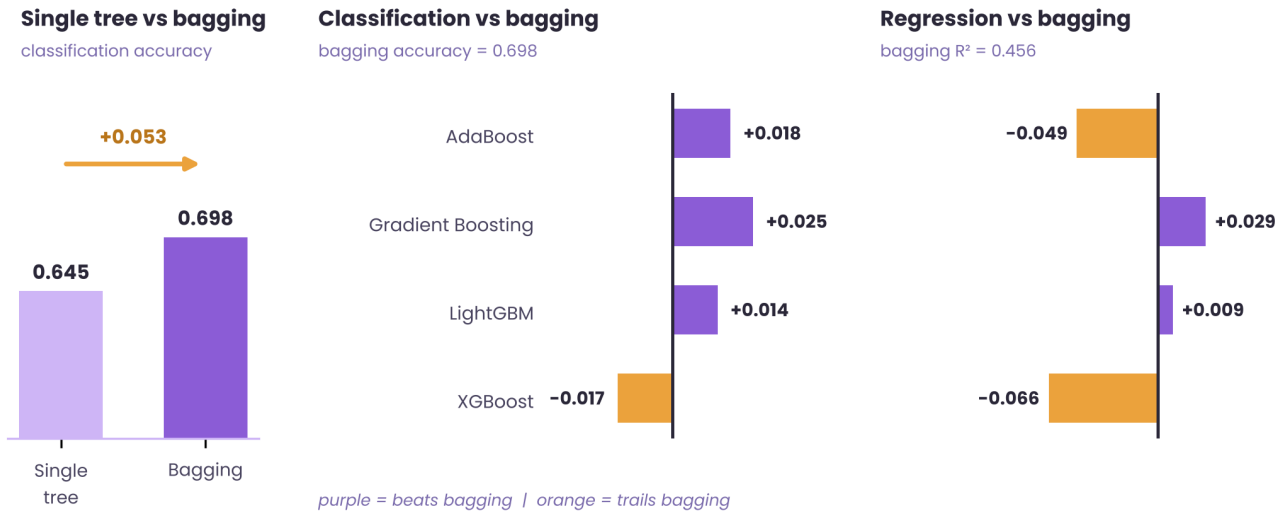


Figure 1: How each boosting model compares with bagging, and the jump from a single tree to bagging

2 Comparison of tree-based boosting algorithms

Each of the four builds trees sequentially. Their test results on both tasks are in the table.

Model	Accuracy	MAE	RMSE	R^2	Note
AdaBoost	0.716	8.57	10.52	0.407	0.656 accuracy with 1 learner, rising to 0.721 at 25
Gradient Boosting	**0.723**	**7.85**	**9.81**	**0.485**	Best on both tasks
LightGBM	0.712	8.03	10.00	0.465	Runner-up regressor; trained in 0.69 s
XGBoost	0.681	8.54	10.68	0.390	Last on both tasks (default settings)

Best model	Gradient Boosting posted the smallest errors (MAE 7.85, RMSE 9.81) and the top scores on both tasks.
Tuning check	For Gradient Boosting, a learning rate of 0.1 with 100 estimators gave the best accuracy (0.723). Learning rates of 0.01 and 0.2 scored 0.711 and 0.712, while 50 and 150 estimators scored 0.717 and 0.719.
XGBoost	With default settings it ranked last on both tasks (accuracy 0.681, R^2 0.390).

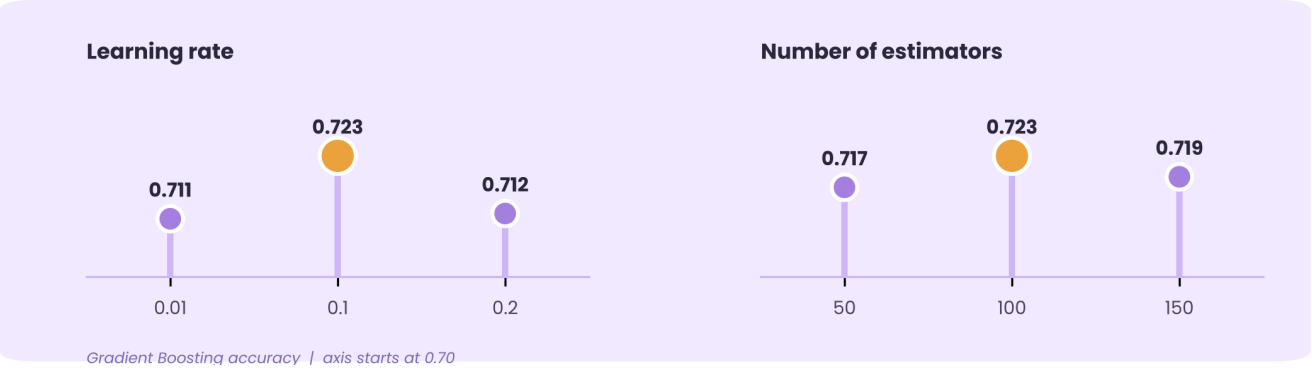


Figure 2: Tuning check for Gradient Boosting, with the best setting highlighted

3 Advantages of voting and stacking

Voting	Simple to use, needs no additional model to train, and blends different model types so their errors differ. Hard voting reached 0.732 accuracy, the best classification result in the notebook; soft voting trailed at 0.679.
Stacking	A meta-learner works out how much weight each base model deserves, and the same idea serves classification and regression. It scored 0.709 accuracy and R^2 0.452.
Trade-off	Both rely on several models, so they are heavier and harder to explain than a single model. Stacking did not surpass Gradient Boosting on either task here.

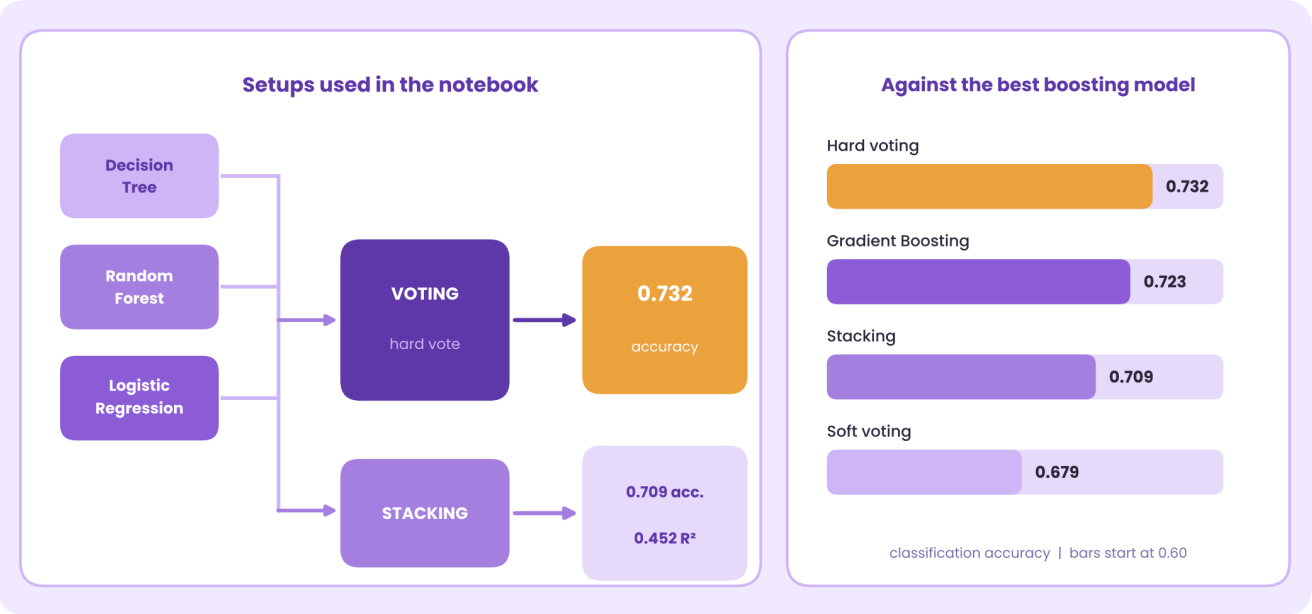


Figure 3: The voting and stacking setups, with their test results beside Gradient Boosting

4 Final model recommendation for deployment

CLASSIFICATION | DEPLOY

Voting Classifier (hard)

Decision Tree + Random Forest + Logistic Regression

Accuracy 0.732

REGRESSION | DEPLOY

Gradient Boosting Regressor

100 estimators, learning rate 0.1 (default)

MAE 7.85 | RMSE 9.81 | R^2 0.485

Why these models	Each posted the best test score for its task in the final comparison (Q30).
Close call	Voting led Gradient Boosting by just 9 students out of 1,040, and hard voting produces no probability scores. Where probabilities are needed, the Gradient Boosting Classifier (0.723) is a near-equal alternative.
Limits	Results come from a single train-test split, and an R^2 of 0.485 leaves about half of the score variation unexplained. Predictions should assist teachers, not replace their judgement.

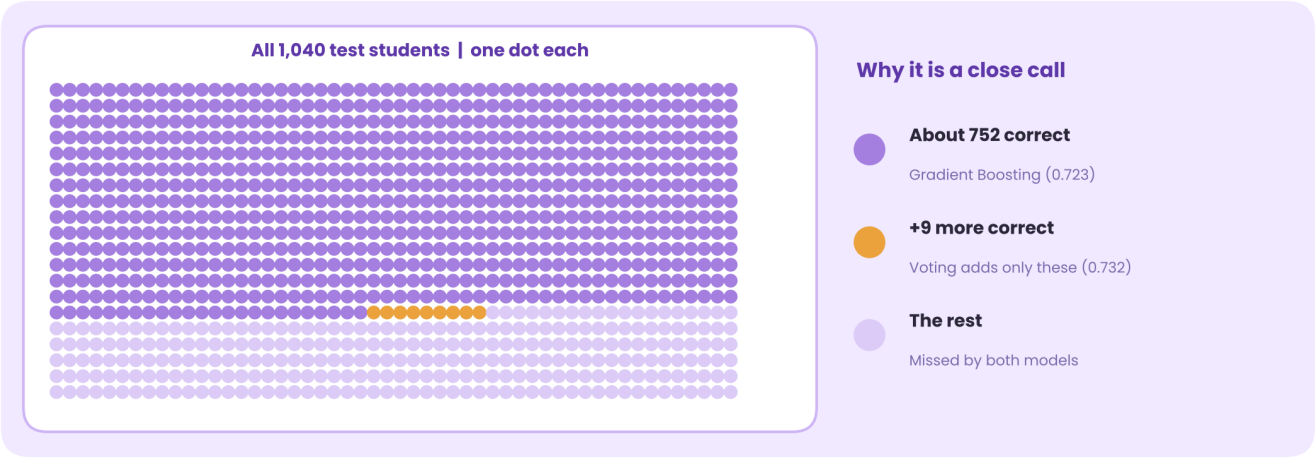


Figure 4: Voting versus Gradient Boosting across all 1,040 test students (approximate counts from the reported accuracies)